

5. Observational Studies

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Roadmap

1. Identification in Observational Studies
2. Selection on Observables
3. Sensitivity Analysis

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 - Rest of the course will cover different designs for observational studies.
- Q: Why are observational studies in causal inference important? (What are the limitations of RCTs?)

Where are we? Where are we going?



Source: Twitter @NobelPrize

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Scientific Background on the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2021

ANSWERING CAUSAL QUESTIONS USING OBSERVATIONAL DATA

The Committee for the Prize in Economic Sciences in Memory of Alfred Nobel

*“Taken together, therefore, the Laureates’ contributions have played a central role in establishing the so-called design-based approach in economics. This approach – aimed at **emulating a randomized experiment to answer a causal question using observational data** – has transformed applied work and improved researchers’ ability to answer causal questions of great importance for economic and social policy using observational data.” (p.2)*

1/ Identification in Observational Studies

Randomized Experiment Review

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- **Randomized experiment** is an experiment with two properties:
 1. **Positivity:** assignment is probabilistic (and not deterministic):
 $0 < \mathbb{P}[D_i = 1] < 1$
 2. **Unconfoundedness:** $\mathbb{P}[D_i = 1 | \mathbf{Y}(1), \mathbf{Y}(0)] = \mathbb{P}[D_i = 1]$
 - Treatment assignment does not depend on any potential outcomes.
 - Sometimes written as $D_i \perp\!\!\!\perp (\mathbf{Y}(1), \mathbf{Y}(0))$.

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- Without unconfoundedness: naive diff-in-means = PATT + selection bias
- **Selection bias:** how different the treated and control groups are in terms of their potential outcome under control.

Selection Bias = Unidentified ATT

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- Example: effect of comment sections on support for online influencers.
 - Naive estimate: influencers do worse without comment sections than with them.
 - $\rightsquigarrow$ negative ATT **OR** positive ATT with large negative selection bias.
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- With an unbounded Y_i , we cannot even bound the ATT because, in principle, SB could be anywhere from $-\infty$ to ∞ .
- We say ATT (as well as ATE) are **unidentified** w/o further assumptions.

What is identification?

- **Identification** connects the counterfactual to the observed.
 - **Counterfactual distribution** $\mathbb{P}^*$ of $\{Y_i(1), Y_i(0), D_i, \mathbf{X}_i\}$.
 - **Observational distribution** $\mathbb{P}$ of $\{Y_i, D_i, \mathbf{X}_i\}$.
 - Causal quantities are functions of $\mathbb{P}^*$, but we get samples from $\mathbb{P}$.
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- Connecting counterfactuals to the observational requires **assumptions**.
 - “**What is your identification strategy?**” = what are the assumptions that allow you to claim that you’ve estimated a causal effect?
 - Research design can help justify assumptions (experiments, RDD, etc).
 - Or you will need to justify them through argument.

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 - $\hat{\tau}_{\text{diff}}$ is an estimator for population diff-in-means
 - Now just estimating conditional expectations, etc.
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 - Purely a statistical question from here on out.
- **Identification** comes first, then comes **estimation**.
 - Without identification, properties of the estimator are unimportant.
 - keep them separate: estimator shouldn't drive identification.

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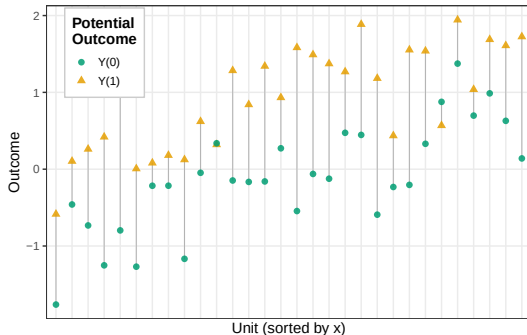
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 - Effect of corporate giants on economic development (confounder: previous economic development)
- Confounding $\rightsquigarrow$ incomplete identification of ATE $\rightsquigarrow$ biased estimators.

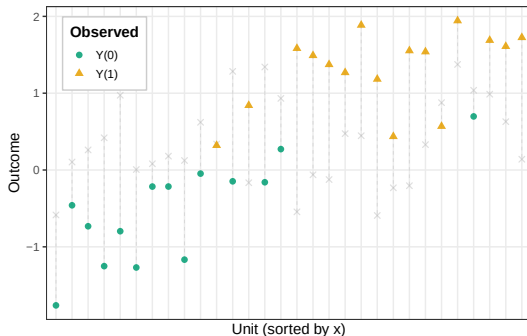
Confounding: The Science Table

- Simulated data (true ATE = 1). Ideally, we observe **both** $Y_i(1)$ and $Y_i(0)$:
 - Each line segment = unit-level treatment effect $Y_i(1) - Y_i(0)$.



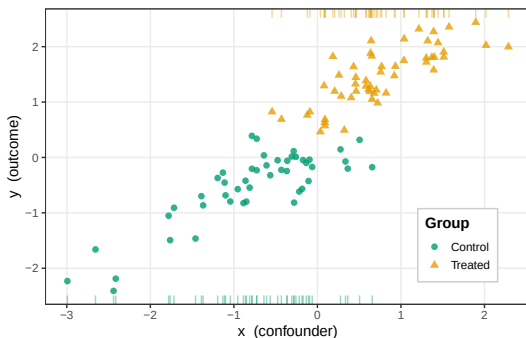
Confounding: The Fundamental Problem

- In reality, we only observe **one** potential outcome per unit.
 - Grey $\times$ = missing counterfactual. We cannot compute individual effects!



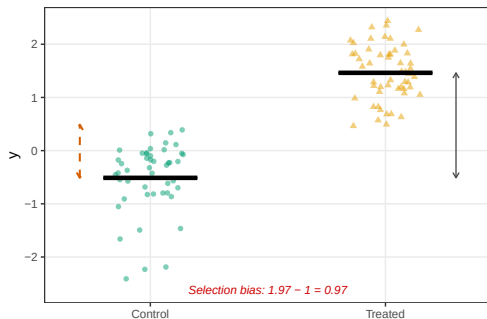
Confounding: What Does It Look Like?

- X_i affects **both** treatment assignment and outcome (confounder).
 - Higher $X_i \rightsquigarrow$ more likely to be treated. Notice the rug plots on top/bottom.
 - (New simulated data with $n = 100$ to visualize the pattern more clearly.)



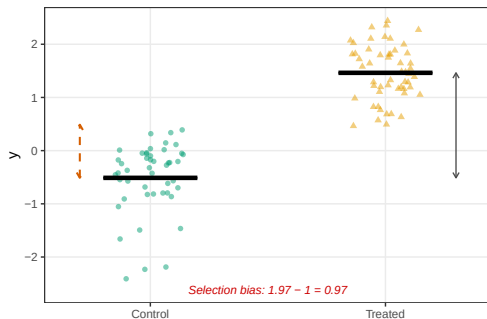
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 - Also called: conditional unconfoundedness, weak ignorability, selection on observables, no omitted variables, exogenous, conditional exchangeability, etc.
 - $\rightsquigarrow$ Conditional on some covariates, D_i is (effectively) randomly assigned.
 2. **Positivity or Overlap:** $0 < \mathbb{P}[D_i = 1 | \mathbf{X}_i] < 1$
 - Treatment and control are both possible at every value of $\mathbf{X}_i$
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- We'll take $\mathbf{X}_i$ as a ‘given’ for now and see later how we might choose it.
- These are assumptions that **can be wrong!**

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- Useful to write the treated and control CEFs:

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- How the mean of the potential outcomes vary with the covariates.
- Key part of the identification above:

$$\underbrace{\mu_1(\mathbf{x})}_{\text{counterfactual}} = \underbrace{\mathbb{E}[\textcolor{red}{Y}_i|D_i = 1, \mathbf{x}_i = \mathbf{x}]}_{\text{observational}}, \quad \mu_0(\mathbf{x}) = \mathbb{E}[\textcolor{red}{Y}_i|D_i = 0, \mathbf{x}_i = \mathbf{x}]$$

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- Procedure:
 1. Obtain predicted values for all units when $D_i = 1$.
 2. Obtain predicted values for all units when $D_i = 0$.
 3. Take the average difference between these predicted values.

Regression and the Imputation Estimator

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- What if we estimate $\widehat{\mu}_1(\mathbf{x})$ and $\widehat{\mu}_0(\mathbf{x})$ jointly in a single linear model?
- Uninteracted OLS: $Y_i = \alpha + \tau D_i + \beta' \mathbf{X}_i + \varepsilon_i$

$$\mathbb{E}[Y_i | D_i = d, \mathbf{X}_i] = \begin{cases} \underbrace{(\alpha + \tau) + \beta' \mathbf{X}_i}_{\mu_1(\mathbf{X}_i)} & \text{if } d = 1 \\ \underbrace{\alpha + \beta' \mathbf{X}_i}_{\mu_0(\mathbf{X}_i)} & \text{if } d = 0 \end{cases}$$

Regression and the Imputation Estimator

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- $\mu_1(\mathbf{X}_i) - \mu_0(\mathbf{X}_i) = \tau$ for all i (same slope β).
- $\rightsquigarrow$ Imputation estimator = coefficient on $D_i = \widehat{\tau}$.

Regression and the Imputation Estimator

- Fully interacted OLS (centered):

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- But on average = τ , since $\tilde{\tilde{\mathbf{X}}} = 0$.
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 - But on average = τ , since $\tilde{\mathbf{X}} = 0$.
 - $\rightsquigarrow$ Imputation estimator = coefficient on $D_i = \hat{\tau}$.
- Different slopes for treated and control $\rightsquigarrow$ equivalent to separate regressions.
 - Key difference: uninteracted assumes **constant** treatment effect; fully interacted allows **heterogeneous** effects.

Variance Estimation

- How do we get estimates of the variance of $\widehat{\tau}_{\text{reg}}$?

Variance Estimation

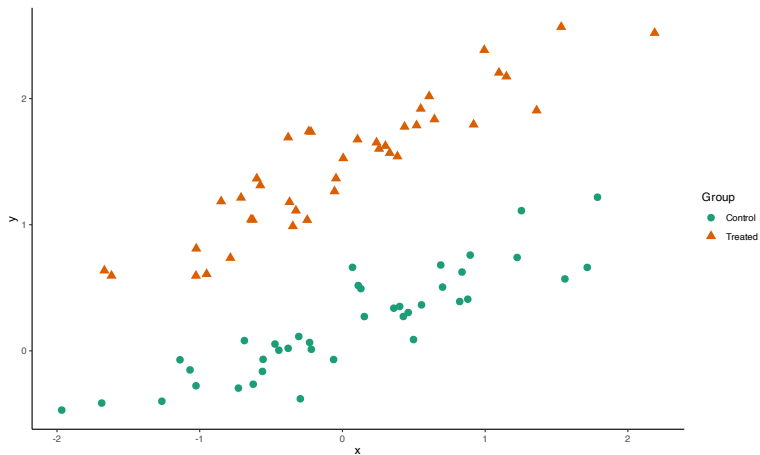
- How do we get estimates of the variance of $\widehat{\tau}_{\text{reg}}$?
- If an OLS coefficient $\rightsquigarrow$ use heteroskedasticity-consistent (robust) variance estimator (e.g., HC2).
- For the general imputation estimator (separate regressions), analytic variance expressions exist but involve estimation uncertainty from both $\widehat{\mu}_1$ and $\widehat{\mu}_0$ — complicated!

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- For the general imputation estimator (separate regressions), analytic variance expressions exist but involve estimation uncertainty from both $\hat{\mu}_1$ and $\hat{\mu}_0$ — complicated!
- Computational alternative: **(nonparametric) bootstrap**
 - Randomly resample n rows of the data with replacement.
 - Refit the regressions on the bootstrapped data.
 - Calculate $\hat{\tau}_{\text{reg}}$ in each bootstrap.
 - Repeat several times and use empirical variance of the bootstraps.

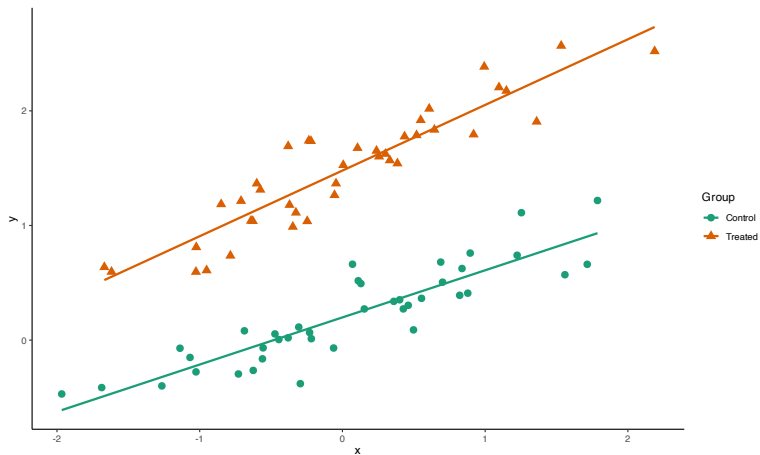
Imputation Estimator Visualization

```
1 > toy_data <- read_csv("https://bit.ly/3v0y2Ao")
```



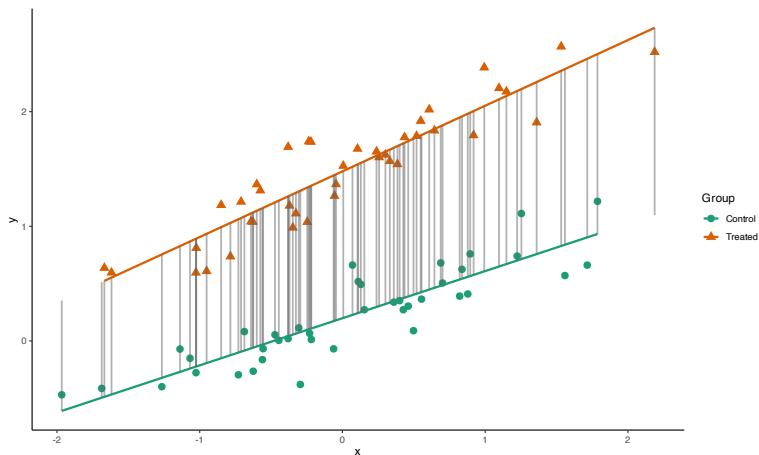
Imputation Estimator Visualization

```
1 > lm0 <- lm(y~x, data = toy_data, subset = d==0); lm1 <- lm(y~x, data = toy_data, subset = d==1)
```



Imputation Estimator Visualization

```
1 > mu0.imps = predict(lm0, toy_data); mu1.imps = predict(lm1, toy_data)
2 > cat("Estimate of ATE:", mean(mu1.imps - mu0.imps))
3 Estimate of ATE: 1.285176
```



Fully Interacted OLS & Imputation Estimator

- Recall: fully interacted OLS with centered covariates = imputation estimator.
- Let's verify numerically:

```
1 > toy_data$x_tilde <- toy_data$x - mean(toy_data$x)
2 > mod_full <- lm(y ~ d + x_tilde + d * x_tilde, data = toy_data)
3
4 > cat("\nEstimate of ATE (Imputation):", mean(mu1.imps - mu0.imps),
5       "\nEstimated coefficient on Di from full int.", mod_full$coefficients["d"])
6
7 Estimate of ATE (Imputation): 1.285176
8 Estimated coefficient on Di from full int. 1.285176
```

- ↪ With centered covariates, $\hat{\tau}_{\text{reg}} \equiv$ estimated coefficient on D_i .
 - Would be the same for uninteracted model, except the variance will be larger (less precision).

Variance Estimation w/ Bootstrap

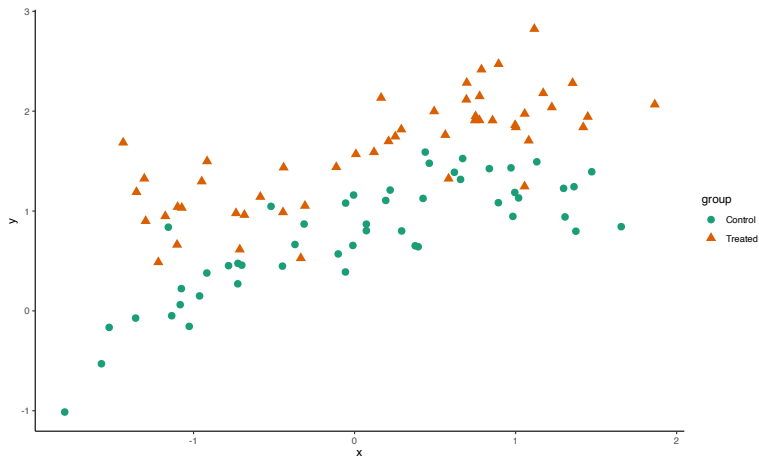
- Again, how do we get estimates of the variance of $\hat{\tau}_{\text{reg}}$?
 - (Nonparametric) bootstrap: a simulation-based alternative to analytic standard errors.
 - Idea: resample from the data to approximate the distribution of $\hat{\tau}_{\text{reg}}$.

```
1 > set.seed(02138); sims<-500; tau_hat_draws<-rep(NA, sims)
2 > for (i in 1:sims) { # Repeat the following several times
3   # 1. Randomly resample n rows of the data with replacement
4   sample_boot <- dplyr::slice_sample(toy_data, n = nrow(toy_data), replace = TRUE)
5
6   # 2. Refit the regressions on the bootstrapped data
7   model <- lm(y ~ d + x_tilde + d*x_tilde, data = toy_data)
8   dat1 <- sample_boot; dat1$d <- 1
9   dat0 <- sample_boot; dat0$d <- 0
10  mu1_hat <- predict(model, newdata = dat1)
11  mu0_hat <- predict(model, newdata = dat0)
12
13  # 3. Calculate tau_hat in each bootstrap
14  tau_hat_draws[i] <- mean(mu1_hat - mu0_hat)
15 }
16
17 > # 4. Use empirical variance of the bootstraps
18 > var(tau_hat_draws)
19 [1] 0.000254049
```

Nonlinear Relationships

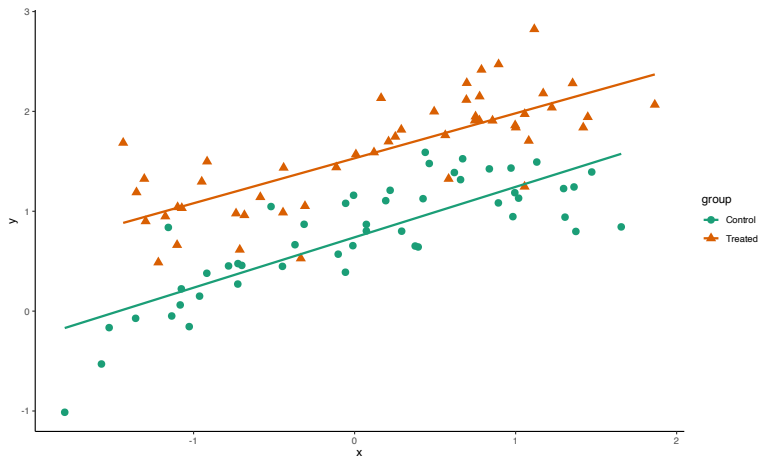
- Same idea but with nonlinear relationship between Y_i and X_i :

```
1 > toy_data_02 <- read_csv("https://bit.ly/4c0g0Pl") # sim data with nonlinear relationship
```



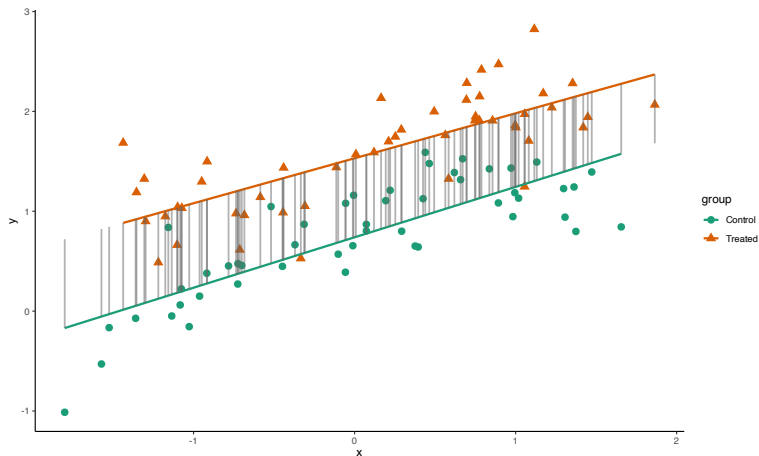
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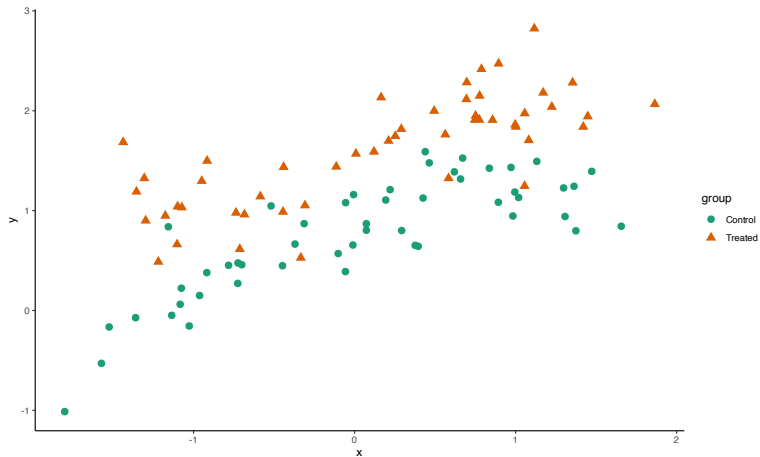


Using Semiparametric Regression

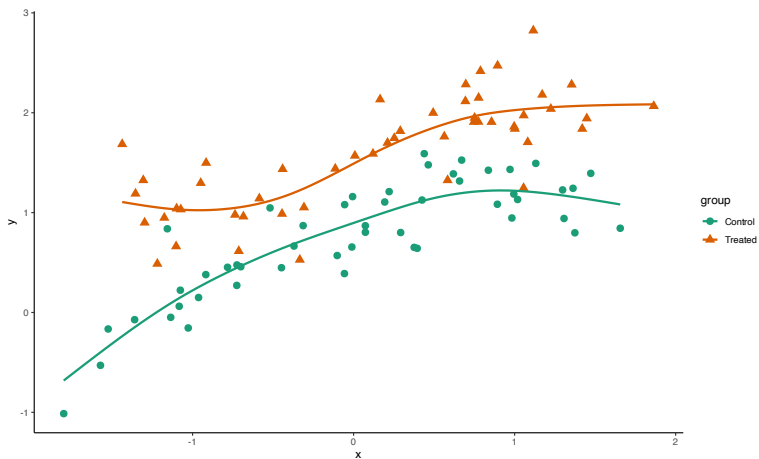
- Here, CEFs are nonlinear, but we don't know their form.
- We can use generalized additive models (GAMs) from the mgcv package for 'flexible' estimation:

```
1 > library(mgcv)
2 > gam1 <- gam(y~s(x), data = toy_data_02, subset = group=="Treated")
3 > gam0 <- gam(y~s(x), data = toy_data_02, subset = d==0); summary(gam1)
4
5 Family: gaussian
6 Link function: identity
7
8 Formula:
9 y ~ s(x)
10
11 Parametric coefficients:
12             Estimate Std. Error t value Pr(>|t|)
13 (Intercept)  1.59527    0.04571   34.9   <2e-16 ***
14 ---
15 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
16
17 Approximate significance of smooth terms:
18             edf Ref.df    F p-value
19 s(x)  3.73   4.642 19.85   <2e-16 ***
20 ---
21 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
22
23 R-sq.(adj) =  0.651   Deviance explained = 67.7%
24 GCV = 0.11539   Scale est. = 0.10447   n = 50
```

Using GAMS



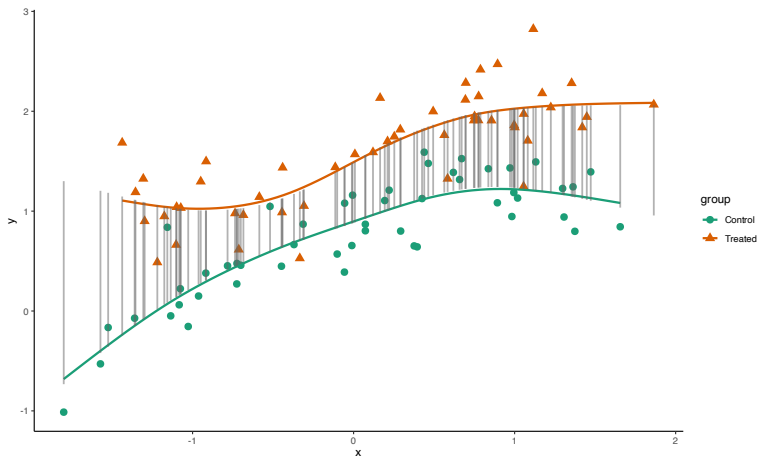
Using GAMS



Using GAMS

- We can estimate $\hat{\tau}_{\text{reg}}$ using the imputation estimator.

```
1 > cat("Estimate of ATE (GAM):",mean(predict(gam1) - predict(gam0)))  
2  
3 Estimate of ATE (GAM): 0.8379884
```



3/ Sensitivity Analysis

Cf. Cinelli & Hazlett (2020), *Making Sense of Sensitivity*, JRSS-B

Sensitivity Analysis for Regression

- Standard regression estimator of the ATE:

$$Y_i = \hat{\alpha} + \hat{\tau}D_i + \mathbf{X}'_i\hat{\beta} + \varepsilon_i$$

- Note: applies equally to fully interacted models (cf. W4).

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- Standard **omitted variable bias (OVB)** formula:

$$\begin{aligned}\hat{\tau} &= \tau + \gamma \times \underbrace{\frac{\text{cov}(D_i^{\perp \mathbf{X}}, U_i^{\perp \mathbf{X}})}{\mathbb{V}(D_i^{\perp \mathbf{X}})}}_{\text{regression of } U_i^{\perp \mathbf{X}} \text{ on } D_i^{\perp \mathbf{X}}} \\ &= \tau + \underbrace{\gamma \times \delta}_{\text{OVB}}\end{aligned}$$

- But γ and δ are coefficients of an **unobserved** variable: we don't know what U is, let alone its scale.

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 - female is binary (0/1), age is continuous (18–80), U is... unknown.
 - If you rescale a variable by 10, its coefficient shrinks by 10.
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 - If you rescale a variable by 10, its coefficient shrinks by 10.
 - $\rightsquigarrow$ Comparing $\gamma = 0.5$ to a coefficient of 0.3 is meaningless if the scales differ.
- Partial R^2** : proportion of residual variance explained by one variable (scale-free).

$$R^2_{Y \sim U | D, \mathbf{X}} = \frac{R^2_{Y \sim D + \mathbf{X} + U} - R^2_{Y \sim D + \mathbf{X}}}{1 - R^2_{Y \sim D + \mathbf{X}}} = \frac{\text{additional variance explained by } U}{\text{variance unexplained by } D, \mathbf{X}}$$

OVb in Terms of Partial R^2

- Cinelli and Hazlett (JRSS-B, 2020) rewrite OVb using partial R^2 :

$$|\text{bias}| = \sqrt{\frac{\overbrace{R^2_{Y \sim U|D, \mathbf{x}}}^{U \rightarrow Y} \cdot \overbrace{R^2_{D \sim U|\mathbf{x}}}^{U \rightarrow D}}{1 - R^2_{D \sim U|\mathbf{x}}} \cdot \frac{\mathbb{V}(Y \perp\!\!\!\perp \mathbf{x}, D)}{\mathbb{V}(D \perp\!\!\!\perp \mathbf{x})}}$$

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- Same structure as $\gamma \times \delta$:
 - $R^2_{Y \sim U|D,\mathbf{x}}$: how much U predicts Y (partialing out $D, \mathbf{x}$) $\rightsquigarrow$ the $U \rightarrow Y$ path.
 - $R^2_{D \sim U|\mathbf{x}}$: how much U predicts D (partialing out $\mathbf{x}$) $\rightsquigarrow$ the $U \rightarrow D$ path.

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 - $1 - R^2_{D \sim U|\mathbf{x}}$: “clean” treatment variation not driven by U . As U dominates D , this $\rightarrow 0$ and bias explodes.
 - $\mathbb{V}(\cdot)/\mathbb{V}(\cdot)$: scales the bias to the same units as $\hat{\tau}$ (computed from data).

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- $\rightsquigarrow$ We can ask: “How strong would U need to be, *relative to observed confounders*, to overturn our finding?”
- **Sensitivity analysis:** vary the two unknown R^2 's $\in [0, 1]$ and trace out how $\hat{\tau}$ changes.
 - This is exactly what the contour plot in the next slides will show.

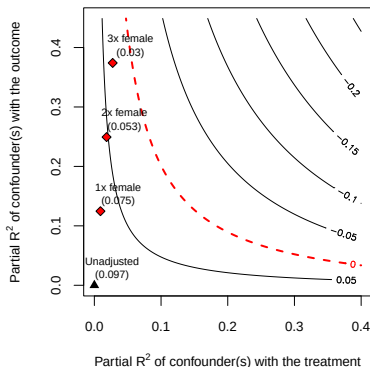
R Package: sensemakr

```
1 # load package
2 library(sensemakr)
3
4 # import dataset
5 data("darfur")
6
7 # run the regression model
8 model <- lm(peacefactor ~ directlyharmed + age + farmer_dar + herder_dar +
9           pastvoted + hhsize_darfur + female + village, data = darfur)
10
11 # conduct the sensitivity analysis
12 sensitivity <- sensemakr(model = model,
13                          treatment = "directlyharmed",
14                          benchmark_covariates = "female",
15                          kd = 1:3)
```

- Darfur conflict (Sudan, 2003) survey: D = directly harmed by violence, Y = support for peacemaking.
- female is the strongest known confounder $\rightsquigarrow$ used as benchmark.
 - kd (knock-on distance): how many times stronger would the unobserved confounder need to be (relative to female) to change our conclusion?

Sensitivity Analysis with Partial R^2

```
1 # plot bias contour of point estimate  
2 plot(sensitivity)
```



- Contour lines = adjusted estimate $\hat{\tau}$.
- Red diamonds = benchmarks at 1–3× strength of female.

- Here: even a confounder 3× as strong as female would not eliminate the effect. $\rightsquigarrow$ Our estimate is robust to plausible levels of unobserved confounding.

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- See also: **5b. Partial Identification** for what we can learn under weaker assumptions (bounds instead of point estimates).
- Up next: DAGs, i.e., a graphical framework for reasoning about which covariates to condition on.

Onto the presentations & discussions!

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